

Protection and Pricing Power

in Sequential Venture Financing

Stephen Dossou

Supervised by : Pierre Fleckinger and Philippe Mustar

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Outline

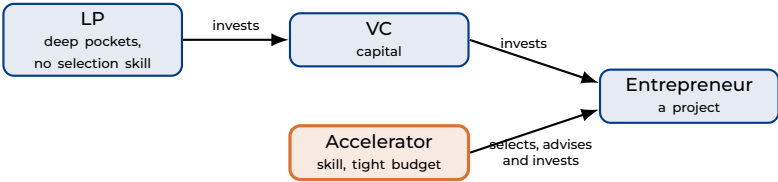
- 1 Introduction
- 2 Literature Review
- 3 Contribution
- 4 Model
- 5 Game 3: the pre-seed investor's own clause
- 6 Taking it to data
- 7 Takeaways

Who does what in venture capital

The classic chain



With an accelerator



Accelerators contracts

The fixed slice is standardised across accelerators. The contingent tranche is not, and it is where the MFN appears.

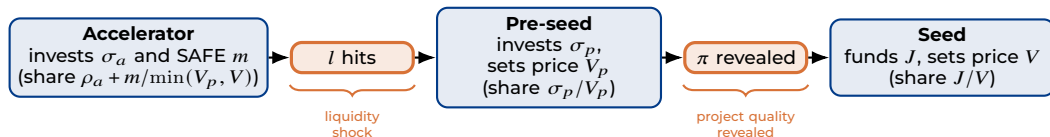
	Fixed amount, fixed share	Contingent tranche
Y Combinator	\$125K for 7%	\$375K uncapped MFN safe, since 2022
Techstars	\$20K for 6% (5% in 2025)	\$100K note capped at \$3M, then MFN in 2025
500 Global	\$150K for 6%	—
Entrepreneur First	\$125K for 8%	\$125K uncapped MFN safe
Antler	£125K for 8.5%	£85K convertible note

Entrepreneur First and Antler: London and UK programmes.

Research question

What does an MFN clause held by the first investor do to the rounds that follow — to the stake each party ends up with, to the amounts they put in, and to the seed probability of projects?

Most Favored Nation explained



- **Sequential contracting.** (McAfee and Schwartz, 1994; Hart and Tirole, 1990)
 - MFN as a commitment remedy — studied for the grantor, here for a third party.
- **Staging and convertibles.** (Gompers, 1995; Kaplan and Strömberg, 2003; Cornelli and Yosha, 2003; Schmidt, 2003)
 - Devices against moral hazard. Here the conversion *basis* is the object.
- **Accelerators.** (Gonzalez-Uribe and Leatherbee, 2017; Hallen et al., 2020)
 - The contract that finances them has not been modelled.
- **Interim finance.** (Admati and Pfleiderer, 1994; Hellmann and Thiele, 2015)
 - Here a claim held by an *earlier* investor, converting on the interim price.

This paper

The accelerator is the only winner of the MFN introduction. The pre-seed investor strictly loses. The probability of seed success and the seed investor's gain are both ambiguous.

Striking findings

- **Accelerator.** Holds $\rho_a^0 + m/\min(V, V_p)$ in place of ρ_a^0 and earns more at every shock, most where its tranche converts lowest.
- **Pre-seed investor.** Earns less at every shock, even where it pre-funds and ends up holding more of the company than in the benchmark.
- **Seed investor.** Same amount, same stake, same payoff where the pre-seed investor advances the shock and no more. Where it pre-funds, the seed investor funds less at a lower threshold, and its payoff has no sign.
- **Projects.** No sign. Fewer where the accelerator's cash covers the shock on its own: the tranche converts at the round and raises the bar by the factor $1 + m/I$. Weakly more where a pre-seed round does occur, and exactly as many where that investor advances the shock and no more.

This paper and YC's claims

Announcing the switch to uncapped MFN in 2022, YC's then-president argued that the new deal terms would:

- **remove the pressure to fundraise on unfavourable terms.**
 - Yes. Fewer firms need a pre-seed round at all, and those that do are priced no lower.
- **make a successful seed round more likely.**
 - Only through the cash. The clause itself takes m off what the pre-seed investor will advance, so companies in that band lose their investor and the round with it; and a company that needs no pre-seed round has nobody to absorb the clause, so its threshold rises. What the cash adds is what decides the sign.

The agents

The project

The round must cover $I \in (0, 1)$: the investment, and the cost c_E the entrepreneur must be left to run the project. Outcome in $\{0, 1\}$ with success probability π , density f . A liquidity shock l hits after the accelerator has invested.

Assumptions

- **(A1)** The average project is worth financing: $\mathbb{E}(\pi - l) - I > 0$ (and f log-concave).
- **(B)** Let β the pre-seed investor's budget: $\beta \leq I$.
- **(C)** The entrepreneur runs the project only if his share covers his cost: $\rho_E \pi \geq c_E$. It is the one participation constraint that survives.

Why (B) is important

Without it the investor funds the round and takes the entire residual equity.

Game 1 (Benchmark)

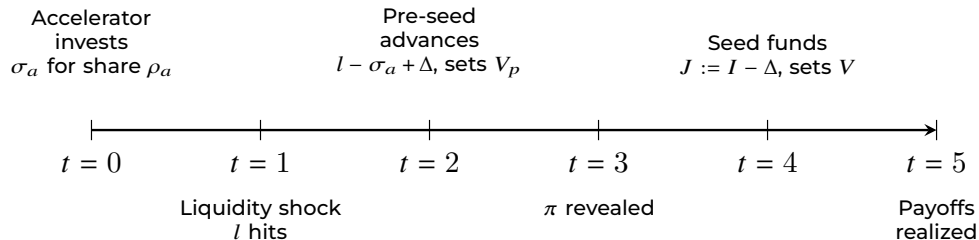


Figure: Game 1 (Benchmark)

Game 1 (Benchmark)

The seed investor maximizes its utility:

$$\max_V \int_{\pi_{\min}}^1 \left(\frac{J}{V(\pi)} \pi - J \right) f(\pi) d\pi$$

Proposition 1

The seed investor
maximizes its utility

$\Leftrightarrow$

Uniform valuation
 $V^* = \frac{J}{1-\rho_a-\rho_p^*}$
(binding the cap table)
The seed finances all
projects above this
valuation

What the round leaves

The entrepreneur keeps $\rho_E^* = c_E/\pi_{\min}$ and the seed investor takes $(J - c_E)/\pi_{\min}$: the round buys the company net of the price of running it.

Game 1 (Benchmark)

The pre-seed investor maximizes its utility:

$$\max_{V_p} \underbrace{\int_{\pi_{\min}}^1 \left(\frac{\sigma_p}{V_p} \pi - \sigma_p \right) f(\pi) d\pi}_{\text{projects which will reach seed}} - \underbrace{\int_0^{\pi_{\min}} \sigma_p f(\pi) d\pi}_{\text{projects which will not reach seed}}$$

Proposition 2

The pre-seed investor
maximizes its utility



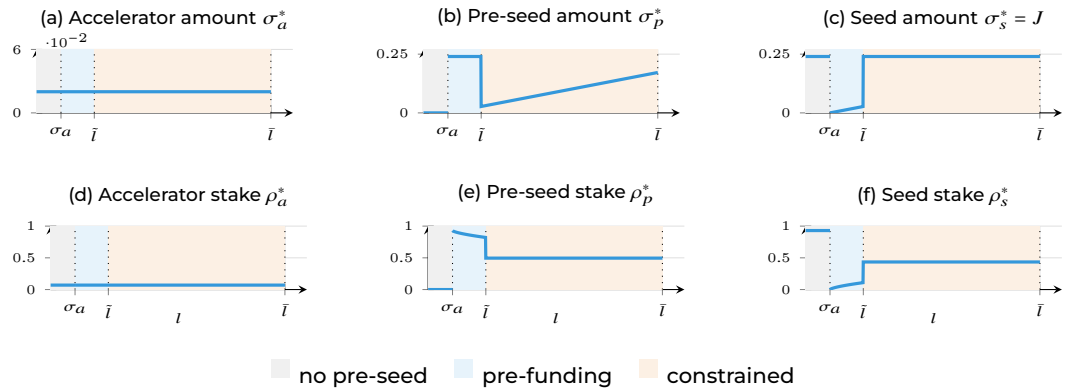
There exists a unique threshold for the pre-seed budget such that:

- **below** only the liquidity shock is covered: $\sigma_p^* = b$
- **above** the budget is saturated, the seed round is pre-funded: $\sigma_p^* = \beta$

The price V_p^* solves the FOC.

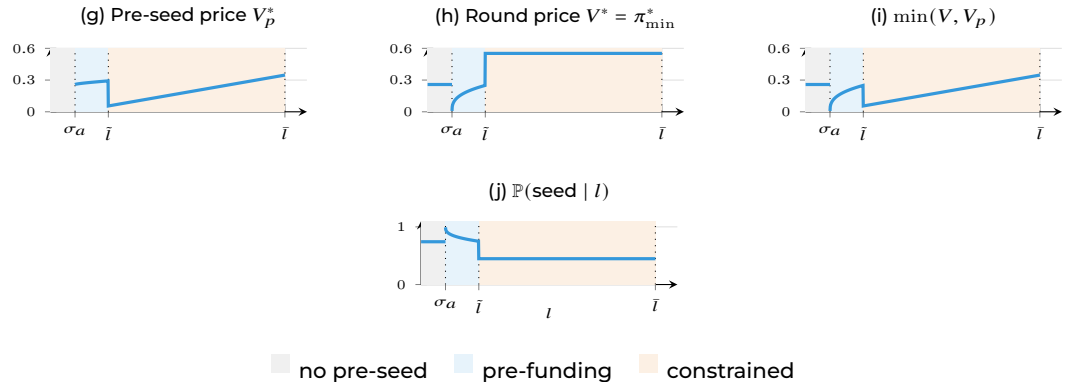
Game 1 (Benchmark)

Below σ_a the pre-seed round is skipped. Between σ_a and $\tilde{l}$ the pre-seed saturates its budget and the seed is left with almost nothing; past $\tilde{l}$ it covers the shock alone and the seed funds J again.



Game 1 (Benchmark)

While it pre-funds, the pre-seed investor prices *above* the round; once constrained, *below* it. The pre-seed investors who pre-fund the seed round increase the seed probability.



Game 2 (MFN)

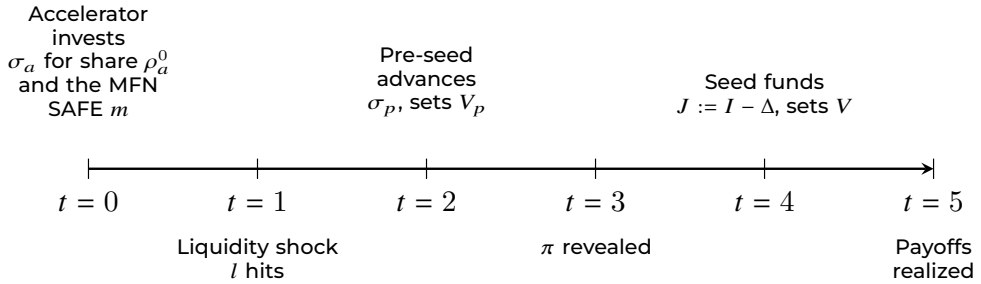


Figure: Game 2 (MFN)

Game 2 (MFN)

V^* and σ_p^* are exactly as in the benchmark.

Proposition 3

There exists a threshold $\bar{\sigma}_p^m$ depending on f, J, m, ρ_a^0 such that the pre-seed investor participates if and only if $\sigma_p^* \leq \bar{\sigma}_p^m$

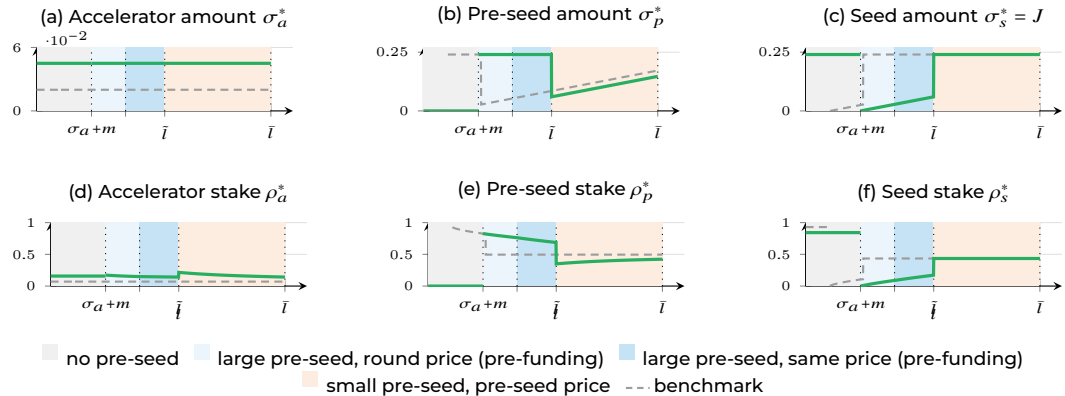
Proposition 4

There exist two thresholds $T_1 < T_2$ depending on f, J, m, ρ_a^0 such that:

- **Small pre-seed, pre-seed price:** $\sigma_p^* \leq T_1$, and $V_p^* < V^*$
- **Large pre-seed, same price:** $T_1 < \sigma_p^* \leq T_2$, and $V_p^* = V^*$
- **Large pre-seed, round price:** $\sigma_p^* > T_2$, and $V_p^* > V^*$

Game 2 (MFN)

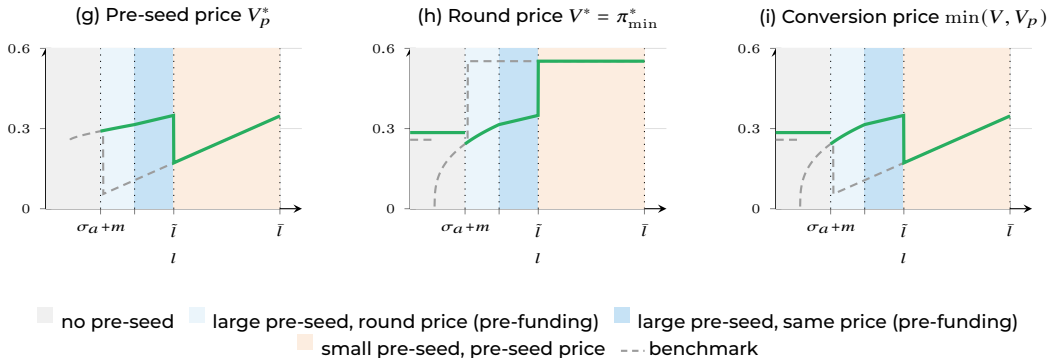
Where the investor pre-funds, σ_p is its budget, not the shock: panel (b) is flat across both large bands, and what separates them is T_2 rising past a fixed amount. Elsewhere ρ_a rises by m/V_p^* and ρ_p falls by the same.



Game 2 (MFN)

“Remove the pressure to fundraise on unfavourable terms.”

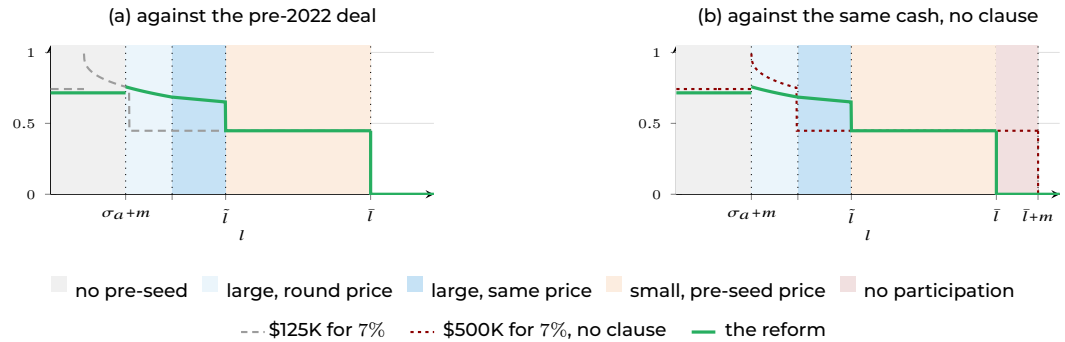
Yes. Fewer firms need a pre-seed round at all, and those that do are priced no lower.



Game 2 (MFN)

“Make a successful seed round more likely.”

Ambiguous. Against the pre-2022 deal, lower where the accelerator’s cash covers the shock on its own and weakly higher wherever a pre-seed round still occurs. Against the same cash without the clause it moves both ways.



Game 2 (MFN)

Proposition 5

In all three regimes, $\Pi_a^{\text{MFN}} \geq \Pi_a^{\text{bench}}$, strict at every $\sigma_p^* < \bar{\sigma}_p^m$.

Proposition 6

In all three regimes, $\Pi_p^{\text{MFN}} \leq \Pi_p^{\text{bench}}$, with equality only in the large pre-seed, round price regime, at a budget that binds in both games.

Game 2 (MFN)

The seed investor funds less and clears a lower threshold. Neither effect dominates.

Proposition 7

$$\Pi_s^{\text{MFN}} - \Pi_s^{\text{bench}} = \underbrace{(J^{\text{MFN}} - J^{\text{bench}}) S(V^*)}_{\leq 0} + \underbrace{J^{\text{bench}} [S(V^*) - S(\pi^{\text{bench}})]}_{\geq 0},$$

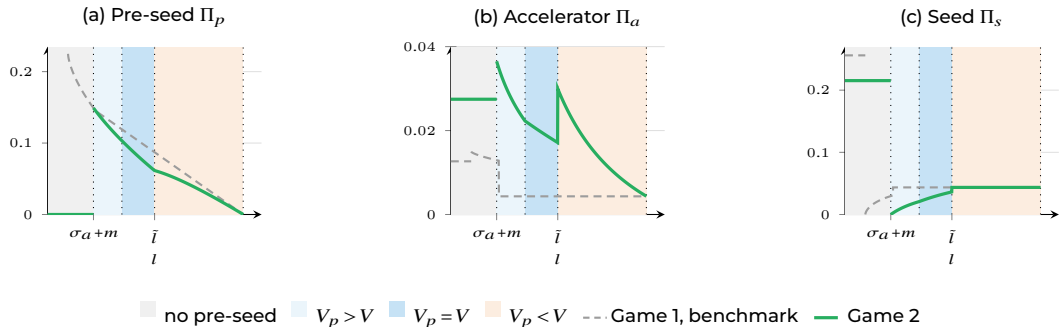
writing $\Pi_s = J S(V)$, $S(V) = M(V)/V - (1 - F(V))$ decreasing. Both terms vanish where the investor advances the shock and no more; the second in the large pre-seed, round price regime at a budget that binds in both games, leaving $-m S(V^*)$.

Corollary

There the seed investor is untouched, and what the accelerator gains the pre-seed investor loses: $\Pi_a^{\text{MFN}} - \Pi_a^{\text{bench}} = -(\Pi_p^{\text{MFN}} - \Pi_p^{\text{bench}}) = \frac{m}{\sigma_p^*} \Pi_p^{\text{bench}}$.

Game 2 (MFN)

The accelerator earns more at every shock, up to +0.030. The pre-seed investor earns less, down to -0.026. The seed investor is untouched past $\tilde{l}$, earns up to 0.041 less before it, and more on the grey band, where the extra cash makes a pre-seed round unnecessary.



Game 3 (capped SAFE)

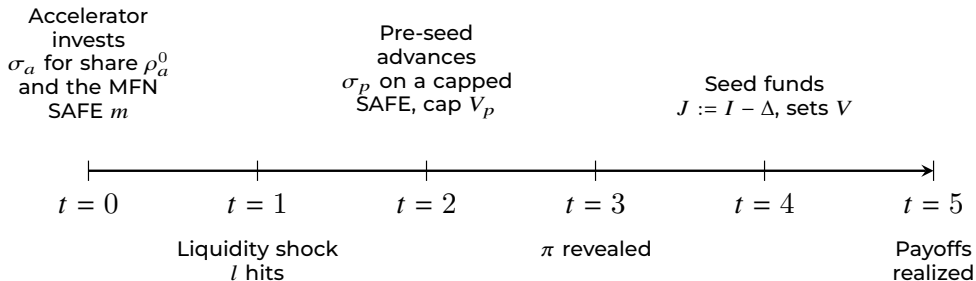


Figure: Game 3 (capped SAFE)

Game 3 (capped SAFE)

Both claims now convert at $\min(V, V_p)$, so $\rho_p = \sigma_p / \min(V, V_p)$.

Proposition 8

T_2 disappears: only one threshold is left, and two regimes instead of three:

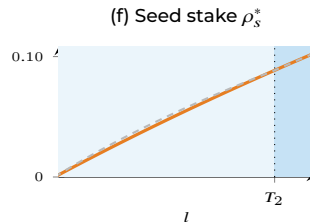
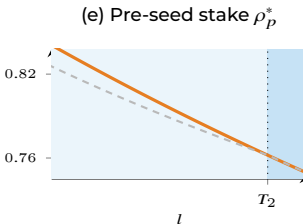
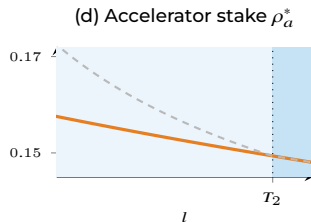
- $\sigma_p^* \leq T_1$: $V_p^* < V^*$, both claims convert at the pre-seed price, threshold π_1
- $\sigma_p^* > T_1$: V_p^* indeterminate, both claims convert at the round price, threshold $\hat{V}_p$

Proposition 9

The pre-seed investor's payoff is never higher under the cap than under a price it sets itself, and strictly lower when $\sigma_p^* > T_2$ — an amount it advances only if $\pi_1 < \bar{x}$.

Game 3 (capped SAFE)

The amounts are identical at every shock, so only the stakes are drawn, and only on the window where the two games differ at all — the band where Game 2's investor prices above the round. Even there, nothing moves by more than 1.5 points.

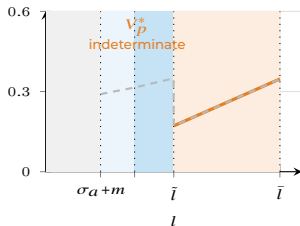


no pre-seed $V_p > V$ $V_p = V$ $V_p < V$ -- Game 2, fixed price — Game 3, capped SAFE

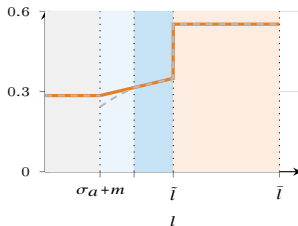
Game 3 (capped SAFE)

Where the cap does not bind, V_p^* is indeterminate: every base above $\widehat{V}_p$ pays the same, so (g) carries no Game 3 curve. The round then prices at $\widehat{V}_p$, above the π_2 Game 2 attains.

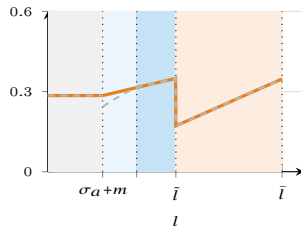
(g) Pre-seed price V_p^*



(h) Round price $V^* = \pi_{\min}^*$



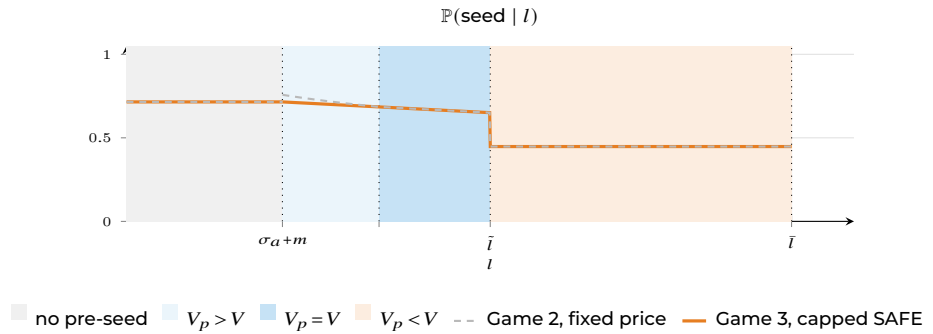
(i) Conversion price $\min(V, V_p)$



no pre-seed
 $V_p > V$
 $V_p = V$
 $V_p < V$
 Game 2, fixed price
 Game 3, capped SAFE

A cap that never binds is a price

The cap costs its holder nothing where it binds — there it is the price the investor would have named anyway — and up to four points of $\mathbb{P}(\text{seed} \mid l)$ where it does not.



Game 3 (capped SAFE)

The two games differ only on $\sigma_p^* > T_2$: the threshold is π_2 when the pre-seed investor sets its own price, and $\widehat{V}_p > \pi_2$ under the cap.

Proposition 10

At a given amount,

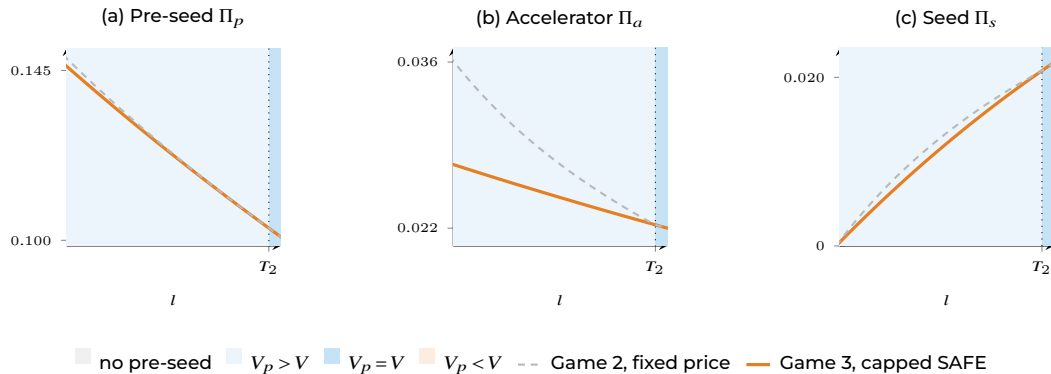
$$\Pi_p^{G3} \leq \Pi_p^{G2}, \quad \Pi_a^{G3} \leq \Pi_a^{G2}, \quad \Pi_s^{G3} \leq \Pi_s^{G2},$$

with equality if and only if $\sigma_p^* \leq T_2$, and all three strict beyond it.

Each payoff is strictly decreasing in the threshold above π_2 : $\Pi_p = \varphi_{J+m}(V)$, which π_2 maximizes; $\Pi_a = \rho_a^0 M(V) + m M(V)/V - \sigma_a - m$; and $\Pi_s = J S(V)$. The table is saturated in both games, so nobody takes what the three give up.

Game 3 (capped SAFE)

Outside this window the two games coincide exactly.
Inside it the cap costs its own holder 1.5% of its payoff at worst, the accelerator 25% and the seed investor 24% — all three closing at T_2 , where the cap stops binding.



The model names the number it cannot supply

The net effect is not signed. What decides it is the interim amount relative to the round it bridges to — so we estimate it.

Setting

- 962 Y Combinator companies, cohorts 2019-H1 to 2021-H2, 24-month window.
- Simulated method of moments, 11 moments: participation, amounts, and their ratio.
- Techstars, deal unchanged 2019–2023, as out-of-sample check.

What the round finances is not only the investment

The model fits, and it fits because I carries the cost of leaving the entrepreneur able to run the project.

		Criterion	d.f.	p
A	$c_E = 0$	35.3	4	$4 \cdot 10^{-7}$
B	$c_E = 0$, budget, entry cost	56.1	2	$7 \cdot 10^{-13}$
C	c_E free — the model	2.20	3	0.53
D	c_E free, budget, entry cost	4.20	1	0.04

$\hat{c}_E = 0.19$, $\hat{\rho}_E = 0.57$, median shock \$4.3M, $\hat{p}_0 = 0.38$.

These firms are in the exact-funding corner

The median interim amount is a fifth of the round that follows: 0.21 observed, 0.22 simulated.

What that settles

- $J \geq J^\dagger$, so the participation cap is $W(J) - m \leq T_1$: no participating investor reaches the first cutoff.
- The price channel is silent — in all 500 bootstrap draws, not as a limiting case.
- What rations is the cap, not the threshold. The clause takes m off what the investor will advance.

The reform moves financing by a tenth of a point

Run through the fitted equilibrium, the cash and the clause nearly cancel, and what is left is negative throughout the confidence interval.

Effect on	Cash	Clause	Total
Pr(priced round), pp	+0.34	-0.53	-0.13
95% interval			[-0.17, -0.02]
Pr(interim round), pp	+2.03	-2.36	-0.33

Seed graduation actually fell 13.8 points over these cohorts. Producing that through the clause would take m at 73.7 times its actual size.

No design on these firms could have found it

A difference-in-differences against Techstars fails its own pre-trend tests — and the estimate explains why rather than excusing it.

Two facts

- Graduation falls $0.282 \rightarrow 0.126$ between 2020-H2 and 2021-H2: half the decline, inside the control period.
- Two thousand firms resolve about a point. The estimand is 0.13.

What reproduces the period is a shift in quality and one in liquidity needs — not the terms.

Main result

The accelerator is the only winner of the MFN introduction. The pre-seed investor strictly loses. The probability of seed success and the seed investor's gain are both ambiguous — and estimated on Y Combinator, the reform moves graduation by -0.13 points.

- **The liquidity shock decides where the tranche converts.** A small one leaves it in the pre-seed block; a large one carries it into the round.
- **Agent by agent.** The accelerator earns more at every shock and the pre-seed investor less. The seed investor is untouched where that investor advances the shock and no more, and its payoff has no sign where it pre-funds.
- **A capped SAFE is worse for the pre-seed investor.** Above T_1 the cap stops binding, the claim converts at V , and every payoff falls.
- **The clause costs financing states even where its price is absorbed.** The cap on the amount falls to $W(J) - m$, so companies in that band lose their investor. Only the accelerator's cash pushes the other way.

Thank you
stephendossou@gmail.com